Customer Shopping Behaviour Analysis

1. Project Overview

This project focuses on analysing customer shopping behaviour using a transactional dataset containing approximately 3,900 purchase records across multiple product categories. The primary objective is to extract meaningful insights that can help businesses better understand how customers interact with products, pricing strategies, and subscription services.

The analysis explores key aspects such as customer demographics (age, gender, and location), purchasing habits, product preferences, and the influence of discounts and promotional offers on buying decisions. Additionally, it examines customer segmentation by identifying patterns among new, returning, and loyal customers to understand retention and lifetime value.

A significant part of the project also investigates the role of subscription programs in driving revenue and customer engagement. By comparing subscriber and non-subscriber behaviour, the project aims to evaluate the effectiveness of subscription-based models.

Using a combination of Python for data preprocessing, SQL for analytical querying, and Power BI for visualization, this project delivers actionable insights that can support data-driven decision-making. These insights can be leveraged to optimize marketing strategies, improve customer retention, enhance product positioning, and ultimately increase overall business performance.

2. Dataset Summary

The dataset used in this project consists of approximately 3,900 transaction records with 18 structured features, providing a comprehensive view of customer purchasing behavior across multiple product categories.

The dataset captures three major dimensions:

- Customer Demographics:
  Includes attributes such as age, gender, location, and subscription status, enabling segmentation and behavioral comparison across different customer groups.
- Purchase Details:
  Covers transactional attributes like item purchased, product category, purchase amount, seasonality, size, and colour. These variables help identify product-level trends and seasonal demand patterns.

- Shopping Behavior Metrics:
  Includes features such as discount usage, promo code application, purchase frequency, previous purchase history, customer review ratings, and shipping preferences. These variables are critical for understanding customer engagement and decision-making drivers.

During data quality assessment, a small number of missing values (37 entries) were identified in the Review Rating column. These were addressed during preprocessing to ensure analytical accuracy.

3. Exploratory Data Analysis & Data Preparation (Python)

Data preprocessing and exploratory analysis were conducted using Python to ensure the dataset was clean, consistent, and analysis-ready.

- **Data Loading & Inspection:**
  The dataset was imported using the pandas library. Initial exploration was performed using functions like df.info() and df.describe() to understand data types, distributions, and summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Handling Missing Values:**
  Missing values in the *Review Rating* column were treated using **median imputation based on product category**, ensuring minimal bias while preserving category-specific rating trends.
- **Data Standardization:**
  Column names were converted into **snake_case format** to maintain consistency and improve readability for further analysis and SQL integration.
- **Feature Engineering:**
  New variables were created to enhance analytical depth:
  - **age_group:** Categorized customers into meaningful age segments for demographic analysis
  - **purchase_frequency_days:** Derived metric to better understand customer buying cycles
- **Data Validation & Cleaning:**
  Redundant features were identified and removed. Specifically, promo_code_used was dropped after confirming overlap with discount_applied, reducing duplication and improving dataset efficiency.
- **Database Integration:**
  The cleaned dataset was exported and loaded into a **PostgreSQL database**, enabling efficient querying and advanced business analysis using SQL.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

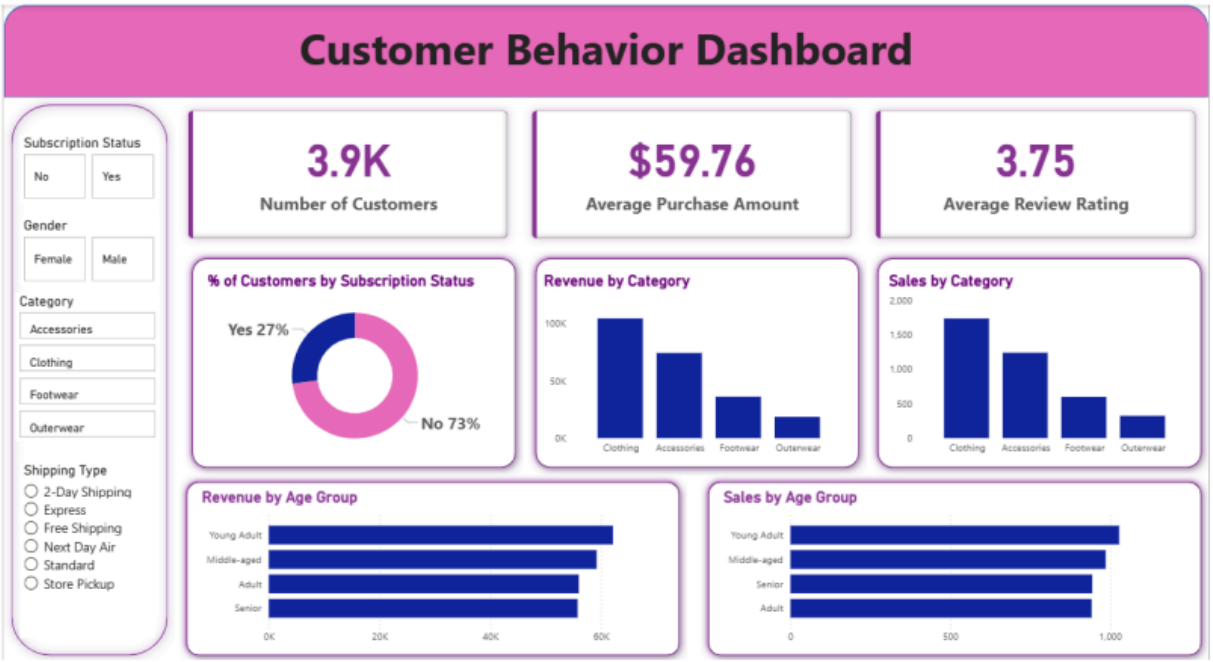
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations (Strategic Insights)

Based on the analysis, the following actionable recommendations are proposed:

- **Enhance Subscription Programs:**
  Introduce exclusive benefits, discounts, or early access offers to encourage more customers to subscribe and improve recurring revenue.
- **Strengthen Customer Loyalty Initiatives:**
  Implement reward systems, personalized offers, and retention campaigns to convert returning customers into loyal, high-value segments.
- **Optimize Discount Strategies:**
  Carefully balance discount usage to drive sales without negatively impacting profit margins, especially for discount-sensitive products.
- **Promote High-Performing Products:**
  Focus marketing efforts on top-rated and best-selling products to maximize conversion rates and customer satisfaction.
- **Adopt Targeted Marketing Approaches:**
  Leverage insights from age groups, spending behavior, and shipping preferences to create personalized campaigns for high-value customer segments.